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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.1

Project ID: R26-DS-017

1. Topic (12 words max)

Multi-Agent LLM Framework for Identifying Fundamental Rights Violations in Sri Lanka

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Access to Fundamental Rights (FR) remedies in Sri Lanka is constrained by language barriers, procedural complexity, and information asymmetry between citizens and legal institutions. Affected users include individual rights-holders, civil society organizations, and frontline institutions such as the Human Rights Commission of Sri Lanka (HRCSL). Many complaints are narrated in Sinhala, Tamil, or mixed-language English, using informal, noisy descriptions that do not align with formal legal schemas. As a result, valid grievances may be delayed, misrouted, or never reach appropriate legal pathways.

The social and legal impact is significant: delayed or failed redress undermines constitutional guarantees (Articles 10-17), weakens trust in institutions, and disproportionately harms marginalized groups. Technically, Sri Lanka lacks tri-lingual, privacy-preserving legal NLP systems tailored to its constitutional framework. Economically, inefficiencies increase institutional workload and legal costs. Ethically and legally, improper handling of personal identifiers risks privacy violations, while inaccurate automated assistance risks misleading users necessitating calibrated uncertainty and human-in-the-loop design.

This research sits within the Foundations of Computing (FoC) research cluster, intersecting Artificial Intelligence, Natural Language Processing, and Information Retrieval, and targets the LegalTech / GovTech industry verticals. It addresses a clear gap: constitution-grounded, citation-backed, multilingual legal AI for the Global South.

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The project contributes to multiple UN Sustainable Development Goals (SDGs):

- SDG 16 (Peace, Justice, and Strong Institutions): by improving access to justice, complaint triage, and precedent transparency.
- SDG 10 (Reduced Inequalities): by enabling Sinhala- and Tamil-speaking citizens equal access to rights mechanisms.

By integrating complaint structuring, rights mapping, precedent retrieval, and admissibility triage with privacy and uncertainty at the core, this research tackles a high-gravity, real-world problem at the intersection of law, language, and computation.

6. Brief description of Existing Research and Systems (300 words max)

Recent advancements in artificial intelligence have led to the development of several systems that support legal reasoning and document analysis. Modern legal AI products and research prototypes increasingly employ large language models (LLMs) and multi-agent systems (MAS) to enhance legal decision support. Systems such as Agents on the Bench simulate collaborative judicial deliberation using multiple agents, improving transparency and reliability in automated legal reasoning (Agents on the Bench, 2024). MASLegalBench provides structured benchmarks to evaluate agent specialization and task decomposition in legal scenarios, particularly in GDPR compliance tasks (MASLegalBench, 2025). Similarly, ChatLaw, a role-based multi-agent legal framework, demonstrates improved legal reasoning accuracy and reduced hallucinations compared to single-agent LLMs (ChatLaw, 2023).

In the Sri Lankan context, existing research has mainly focused on legal information retrieval rather than active legal assistance. An integrated system has been proposed to enhance the retrieval of Sri Lankan Supreme Court verdicts using custom Named Entity Recognition (NER), legal knowledge graphs, and BART-based document summarization (An Integrated Approach to Enhance Legal Information Retrieval of Sri Lankan Supreme Court Verdicts, 2023). These systems improve case search efficiency and document comprehension for legal professionals by organizing judgments into structured and searchable formats.

However, several limitations remain. Existing Sri Lankan systems are largely passive, requiring users to manually interpret constitutional provisions and legal procedures. There is no AI system tailored to Sri Lanka's Fundamental Rights (FR) framework under Articles 10-17, nor support for complaint structuring, rights mapping, or procedural validation. Additionally, Sri Lanka's tri-lingual legal environment (Sinhala, Tamil, English) and privacy-sensitive handling of constitutional complaints are insufficiently addressed. These gaps highlight the need for a domain-specific, multilingual, multi-agent legal assistance system focused on citizen-oriented Fundamental Rights support.

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References

[1] X. Y. Cong Jiang, "Agents on the Bench: Large Language Model Based Multi Agent Framework for Trustworthy Digital Justice," 2024. [Online]. Available: https://arxiv.org/abs/2412.18697.

[2] Huihao Jing, Wenbin Hu, Hongyu Luo, Jianhui Yang, Wei Fan, Haoran Li, Yangqiu Song, "MASLegalBench: Benchmarking Multi-Agent Systems in Deductive Legal Reasoning," 2025. [Online]. Available: https://openreview.net/forum?id=GIzoGvVD3w.

[3] Jiaxi Cui, Munan Ning, Zongjian Li, Bohua Chen, Yang Yan, Hao Li, Bin Ling, Yonghong Tian, Li Yuan, " A multi-agent collaborative legal assistant with knowledge graph enhanced mixture-of-experts large language model," 2023. [Online]. Available: https://arxiv.org/abs/2306.16092.

[4] Shengbin Yue, Ting Huang, Zheng Jia, Siyuan Wang, Shujun Liu, Yun Song, Xuanjing Huang, Zhongyu Wei, "Multi-Agent Simulator Drives Language Models for Legal Intensive Interaction," 2025. [Online]. Available: https://arxiv.org/abs/2502.06882.

[5] Taicheng Guo, Xiuying Chen, Yaqi Wang, Ruidi Chang, Shichao Pei, Nitesh V. Chawla, Olaf Wiest, Xiangliang Zhang, "Large Language Model based Multi-Agents: A Survey of Progress and Challenges," 2024. [Online]. Available: https://arxiv.org/abs/2402.01680.

[6] Xiang Li; Xiaolong Zhou; Jinbo Li; Qingyue Di; Lei Jing; Bin Fan, "Retrieval-Augmented LLM-Driven Multi-Agent Optimization Framework for Intelligent Manufacturing Scheduling," 2025. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/11207301.

[7] Mario De Jesus; Perfect Sylvester; William Clifford; Aaron Perez; Palden Lama, "LLM-Based Multi-Agent Framework for Troubleshooting Distributed Systems," 2025. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/11108224.

[8] V. Gogineni, "LLM-Powered Multi-Agent Systems: A Technical Framework for Collaborative Intelligence Through Optimized Knowledge Retrieval and Communication," 2025. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/11077480.

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7. Brief description of the solution's nature, including a conceptual diagram(maximum 500 words).

Our proposed system addresses the gaps identified in Section 6 by providing a domain-specific, multilingual, privacy-conscious, and procedural-aware legal AI framework for supporting Fundamental Rights (FR) in Sri Lanka. Existing legal AI research and systems, both globally and locally, primarily focus on information retrieval, monolingual document analysis, or passive assistance. In contrast, no system actively supports tri-lingual complaint structuring, constitutional article mapping, citation-grounded precedent retrieval, admissibility checking, or referral guidance for citizen-oriented FR complaints.

System Perspective: (Input → Process → Output)

Input: Free-text FR complaints in Sinhala, Tamil, and English

Case metadata (dates, parties, prior petitions)

Process:

Component 1 - Multilingual Complaint Structuring + Privacy:

Extracts structured entities (plaintiff, defendant, grievance type) and relationships from noisy multilingual narratives while performing de-identification to preserve privacy. This fills the gap in multilingual, privacy-preserving legal information extraction.

Component 2 - Fundamental Rights Article Mapper:

Performs multi-label classification linking complaint content to relevant constitutional articles (10-17), generating evidence-based rationales and calibrated confidence scores. This addresses the lack of structured article mapping with explainable reasoning in Sri Lankan FR systems.

Component 3 - Precedent Retrieval + Citation-backed Output:

Retrieves relevant past cases, statutes, and judicial interpretations, enforcing "no citation, no claim" rules to minimize hallucinations and increase reliability. This mitigates the gap in trustable precedent retrieval with literal citation grounding.

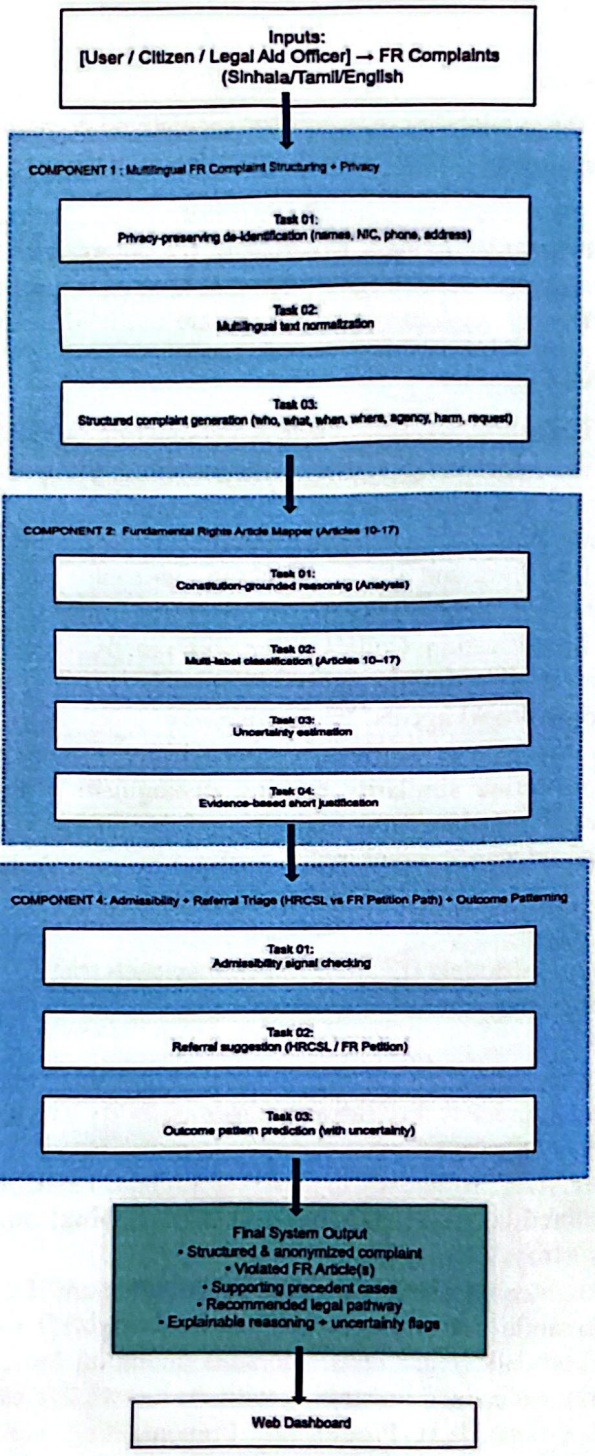
Component 4 - Admissibility + Referral Triage + Outcome Patterning:

Checks basic admissibility criteria, recommends the appropriate referral path (Human Rights Commission vs FR petition), and predicts coarse outcome patterns with uncertainty quantification. This addresses the absence of procedural guidance and triage in current legal AI models.

Output: Structured complaint schema (entities + relationships)
Article mapping with rationale and confidence scores.

Citation-backed precedent summaries. Admissibility status, referral path recommendation, and predicted outcome patterns.

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8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

This project requires interdisciplinary expertise at the intersection of constitutional law, artificial intelligence, and multilingual natural language processing. The required expertise, data sources, and ethical considerations are outlined below.

Specialized Domain Expertise

- Sri Lankan Constitutional Law: Deep understanding of Fundamental Rights under Articles 10-17, including rights interpretation, admissibility conditions, and Supreme Court reasoning patterns.
- Fundamental Rights Procedures: Familiarity with FR petition workflows, procedural timelines, and alternative complaint routes such as the Human Rights Commission of Sri Lanka (HRCSL) to inform complaint triage logic.
- Legal Rule Identification: Collaboration with practicing lawyers and academics to identify admissibility criteria and constitutional reasoning patterns, validating the "rules" used by the AI agents.
- Legal Case Analysis & Retrieval Logic: Ability to analyze Supreme Court FR judgments to define similarity metrics, distinguishing between fact-based and principle-based matches, and optimizing RAG retrieval performance through chunking and relevance reranking.

Technical Knowledge and Skills

- Large Language Models (LLMs): Experience with transformer-based models for text understanding, multi-label classification (Article mapping), and rationale generation.
- Multi-Agent Systems: Designing role-based agents for complaint structuring, rights mapping, precedent retrieval, and admissibility triage, including consistency checks between agents.
- Multilingual NLP: Proficiency in handling Sinhala, Tamil, and English legal texts, including mixed-language narratives (code-switching) and ensuring translation consistency across languages.
- Retrieval-Augmented Generation (RAG) Architecture: Designing pipelines that combine semantic embeddings with keyword (hybrid) search to ground LLM outputs in authentic legal texts. Includes chunking long judgments, reranking results for relevance, and integrating with the knowledge base.
- Privacy-Preserving Text Processing: Implementing automatic redaction (PII masking) to remove names, NICs, and phone numbers from complaint narratives before processing.
- AI Tooling: Familiarity with frameworks such as Hugging Face Transformers, spaCy, vector databases (e.g., Pinecone, Chroma), and Python-based NLP pipelines.

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Data Requirements and Availability

- Fundamental Rights Judgments: Publicly available Supreme Court FR case judgments (PDF/HTML) obtained from official sources, serving dual purposes:
- Training Data: "Facts/background" sections used to train complaint structuring and classification models.
- Knowledge Base: Full texts processed into a vector knowledge base to support RAG precedent retrieval.
- Complaint Form Schemas: Public complaint forms (from HRCSL or Police) used to define structured output fields (e.g., Date, Location, Officer Name).
- Constitutional Text: The Constitution of Sri Lanka (Articles 10-17) defines the static label space for rights mapping.
- Procedural Guidelines: HRCSL complaint-handling documentation and SOPs to support admissibility checks and referral triage logic.
- Law Reports & Archives: Public law report repositories expand precedent coverage beyond recent website uploads.
- Synthetic Data Generation: Paraphrased and translated public case facts create cross-lingual training examples and balance rare outcome classes.

Ethical Considerations

- Privacy Protection: Automatic anonymization and redaction of personal identifiers in complaint narratives to ensure user safety.
- Responsible AI Use: System outputs are explicitly framed as "preliminary decision support," not legal advice, with clear uncertainty indicators (e.g., "Human Review Recommended").
- Legal Oversight: Continuous lawyer involvement ensures responsible interpretation of legal rules and procedural logic.

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9. Objectives and Novelty

Main Objective To develop a domain-specific Multi-Agent LLM Framework for the Sri Lankan legal context that identifies Fundamental Rights violations (Articles 10–17) by automating the transformation of multilingual, free-text complaints into privacy-preserved structured records, mapping incidents to constitutional articles with evidence-based justifications, retrieving citation-backed precedents, and performing admissibility triage to recommend appropriate legal pathways.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Bandara W A P S IT22245960	Multilingual FR Complaint Structuring + Privacy	Convert Sinhala/Tamil/English free-text incident stories into a standard structured complaint record and privacy-preserving de identification.	Sri Lanka-specific tri-lingual complaint-to-structure modeling + privacy-preserving de identification tuned for noisy, mixed-language, real-world narratives.
Nethmini R A I IT22265906	Fundamental Rights Article Mapper (Articles 10-17)	A multi-label classifier + short rationale generator that maps an incident to likely violated Fundamental Rights Articles (10-17).	Constitution-grounded predictions: outputs must include (1) Article number(s), (2) brief evidence-based justification, and (3) calibrated uncertainty ("needs human review" when unsure).
Withanage W W D T H IT22223012	Precedent Retrieval Engine (FR Case Similarity) + Citation Output	A retrieval system that finds the most similar FR cases for a new incident and produces a short explanation that is strictly backed by retrieved case text (citation-backed output).	Sri Lanka-specific legal semantic retrieval + reranking optimized for FR disputes, plus a "no citation, no-claim" generation rule to reduce hallucinations.

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Karunaarachchi K A I H IT22022974	Admissibility + Referral Triage (HRCSL vs FR Petition Path) + Outcome Patterning	A triage model that: (1) checks basic admissibility signals and missing information, (2) suggests an appropriate route (e.g., human-rights complaint workflow vs FR petition pathway), (3) predicts coarse outcome patterns with uncertainty (not legal advice).	Joint modeling of (a) procedural triage, (b) route selection, and (c) outcome pattern prediction, with explicit uncertainty + consistency checks across agents.

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10. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	

b) The proposed topic exhibits novelty.

Yes		No	

c) The student group can successfully execute the proposed project.

Yes		No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Samadhi	Rathnayake	
Co-Supervisor	Ms.	Adya	Dissanayake.	
C External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

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